

One Bit Positions

Time limit: 1.00 s

Memory limit: 512 MB

You are given a binary string of length n . Your task is to calculate, for every k between $1 \dots n - 1$, the number of ways we can choose two positions i and j such that $i - j = k$ and there is a one-bit at both positions.

Input

The only input line has a string that consists only of characters 0 and 1.

Output

For every distance k between $1 \dots n - 1$ print the number of ways we can choose two such positions.

Constraints

- $2 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
1001011010	1 2 3 0 2 1 0 1 0